

Essentials of Healthcare Supply Chain Management

Eyerusalem B/Meskel (B.Pharm, MSc)

Addis Ababa University

School of Pharmacy

Department of Pharmaceutics and Social Pharmacy

Objective

- **Fundamentals of Healthcare Supply Chain Management (SCM)**, focusing on key terminology, strategies, and global trends.
- The essential **concepts** and **terminology** in SCM, the role of **globalization** in healthcare supply chains, and the **critical impact** of SCM on global health outcomes.
- **Supply chain challenges**, including drug shortages, counterfeit medicines, and strategies to ensure **security and sustainability**.

Introduction

- The healthcare supply chain is the backbone of modern healthcare systems, ensuring that **Pharmaceuticals** are available when and where they are needed.
- Effective SCM **improves patient outcomes** by ensuring timely delivery of **Pharmaceuticals**, while also controlling **costs** and reducing **waste**.
- As healthcare becomes increasingly **globalized**, SCM plays a central role in **access to healthcare** and the **achievement of global health goals**.
- Challenges such as **global supply disruptions**, **product quality** (e.g., counterfeit medicines), and **inefficient logistics** require innovative solutions to protect **patient safety** and ensure **equitable access to care**.

Why Pharmaceutical supply management (PSM)?

- Lack of access to medication
- Irrational drug use

DSM is needed as it is concerned with practical ways in which high quality essential drugs are available, affordable and used rationally.

What is Supply Chain Management?

- **Definition:**

A **supply chain** is a network of organizations, people, activities, information, and resources involved in the **production and delivery** of a product or service from the initial **raw materials** to the **final consumer**.

- end-to-end process

Healthcare Supply Chain Management

- In healthcare, the supply chain encompasses all the stages involved in delivering **medical products, supplies, and pharmaceuticals** to **patients**.
- This includes:
 - Manufacturers of **medicines, medical devices, and health equipment**.
 - Distributors and wholesalers.
 - **Pharmacies, hospitals, clinics, and healthcare providers**.
 - **Patients** who receive medical care and products.

- **Manufacturers → Distributors → Hospitals/Pharmacies → Patient**

Goal in Healthcare:

- The primary goal of SCM in healthcare is to ensure that medical products, such as **drugs**, **surgical tools**, and **medical equipment**, are available to healthcare providers at the **right time**, in the **right quantity**, and at the **right cost**.
 - **Timeliness**: Ensuring products arrive when needed.
 - **Quality**: Ensuring products are safe, effective, and compliant with regulations.
 - **Cost-effectiveness**: Managing costs to keep healthcare affordable.

Key Objectives of Healthcare Supply Chain Management

\$Cost Control:



- Minimizing costs through effective procurement, inventory management, and logistics while ensuring that product quality and patient safety are not compromised.
- Strategies include bulk purchasing, vendor management, and optimizing delivery routes.

2 Quality Assurance:



- Ensuring that medical products meet **regulatory standards** and are **safe** for use by patients.
- Includes quality checks at every stage of the supply chain, from manufacturing to delivery.

3 Availability of Products:



- Ensuring that **critical medical supplies** (e.g., medications, vaccines, surgical tools) are available when needed, without **delays** or shortages.
- Efficient **inventory management** and **forecasting** of demand are key to avoiding stockouts.

4 Efficiency in Logistics and Distribution:



- Streamlining the transportation and delivery processes to reduce delays and ensure that products are delivered at the **right place and time**.
- Includes route optimization, warehouse management, and reducing lead times.

Key Supply Chain Terminology and Concepts

Suppliers:

- Entities that provide raw materials, components, or products that are used in the production process.
- It can include **pharmaceutical manufacturers, medical device producers, or raw material suppliers.**

Distributors:

- Organizations or companies that handle the transportation, warehousing, and delivery of products from manufacturers to retailers or healthcare providers.
- **Logistics:**
The management and coordination of the movement of goods, from point of origin to point of consumption.

Key Supply Chain Terminology and Concepts

- **Inventory:**

It refers to the medicines, medical devices, and equipment held in storage (The stock) by hospitals, pharmacies, and distributors.

- **Lead Time:**

The amount of time between the initiation of a process (e.g., an order) and its completion (e.g., delivery).

Key Supply Chain Terminology and Concepts

- **Push Strategy:**
 - Products are produced in advance based on **forecasted demand** and then pushed through the supply chain to the market. This strategy often relies on **centralized distribution systems** and **inventory holding** to ensure products are available when needed.
 - **Example** : Hospitals and clinics might stockpile **common medications** or **medical supplies** based on predicted seasonal demand (e.g., flu season).
- **Pull Strategy:**
 - Products are only produced or ordered when there is actual demand, minimizing inventory holding. It requires close coordination between suppliers and customers and relies on a **just-in-time inventory** model.
 - **Example** : A **pharmacy** orders medications from a supplier only when a prescription is made, rather than holding large stocks in inventory.

Key Supply Chain Terminology and Concepts

- **Postponement Strategy**

A strategy where the final **customization** or **assembly** of a product is delayed until customer demand is more certain. This helps reduce the risk of overstocking or producing excess inventory based on uncertain demand forecasts.

- **Example in Healthcare:**

- **Final packaging of medicines:** Pharmaceutical companies may delay the final packaging or labeling of medicines until the product's destination is confirmed, allowing them to adjust to **regional preferences** or **regulatory requirements**.
- **Medical Devices:** Some medical devices may be assembled in a generic form and customized based on the needs of a specific healthcare facility.

Key Supply Chain Terminology and Concepts

- **The Bullwhip Effect**
- The **Bullwhip Effect** occurs when **small fluctuations in demand** at the consumer level cause **larger fluctuations** up the supply chain. This often leads to **overproduction**, **excess inventory**, or **shortages** as companies try to compensate for these fluctuations.
- **Example in Healthcare:**
 - A **hospital** might order **extra supplies** (e.g., surgical gloves or masks) after hearing about a potential shortage in a nearby facility, causing a **larger-than-necessary order** to be placed with the manufacturer.
 - **Over-ordering** by distributors, based on inaccurate demand forecasts, may cause **disruptions in supply**, leading to **waste** or **shortages** elsewhere.

Key Supply Chain Terminology and Concepts

- **Strategic Alliances**

-

A **strategic alliance** is a formal collaboration between two or more organizations to achieve a common goal, typically to **improve efficiencies**, reduce costs, or expand market reach.

- **Example in Healthcare:**

- **Partnerships between pharmaceutical manufacturers and hospitals** to ensure that critical medicines are available during shortages.

- **Collaborations between medical device companies and distributors** to ensure timely delivery and reduce costs associated with inventory and logistics.

- These alliances often lead to better **information sharing, resource pooling, and risk mitigation.**

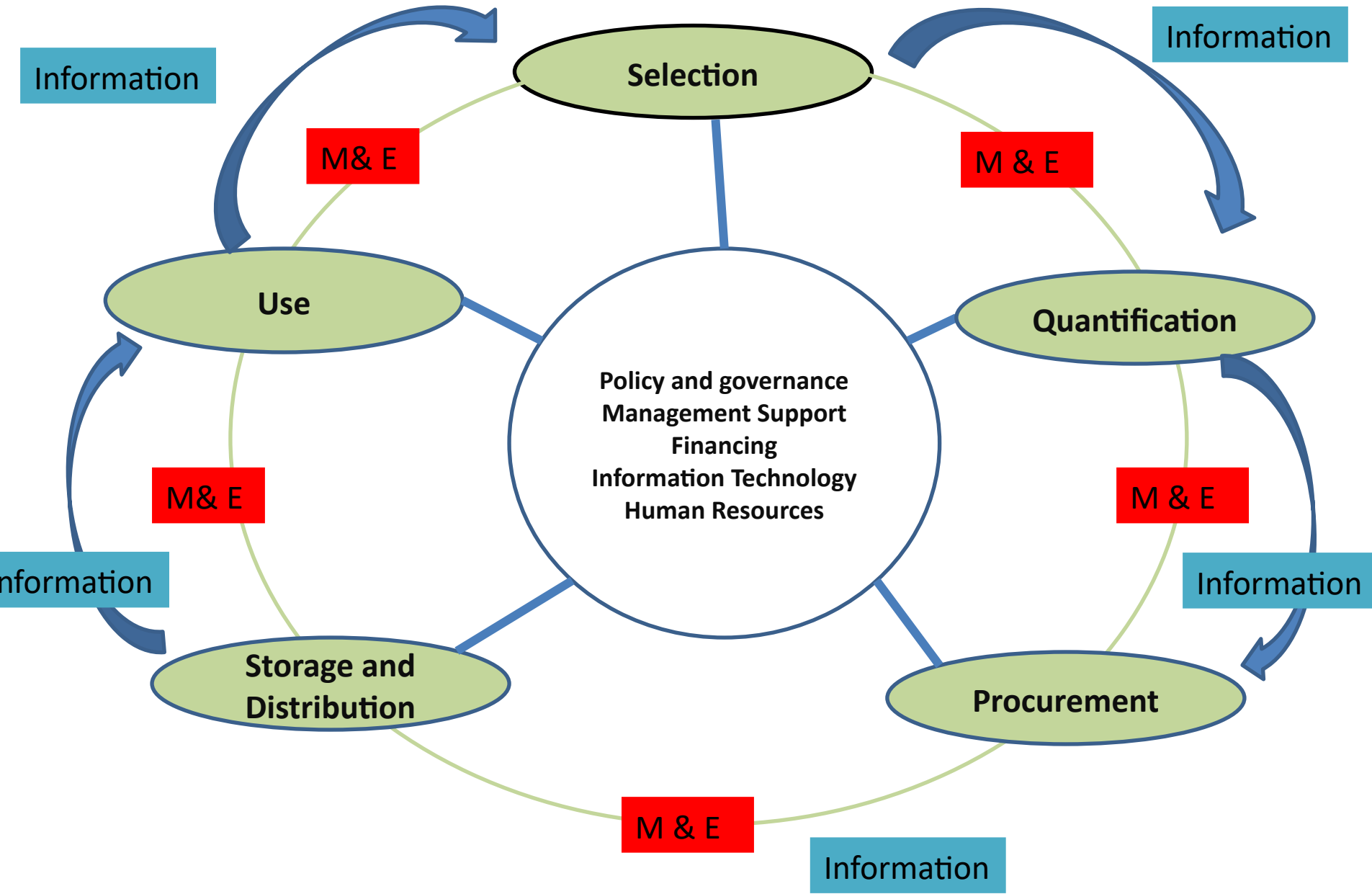
Key Supply Chain Terminology and Concepts

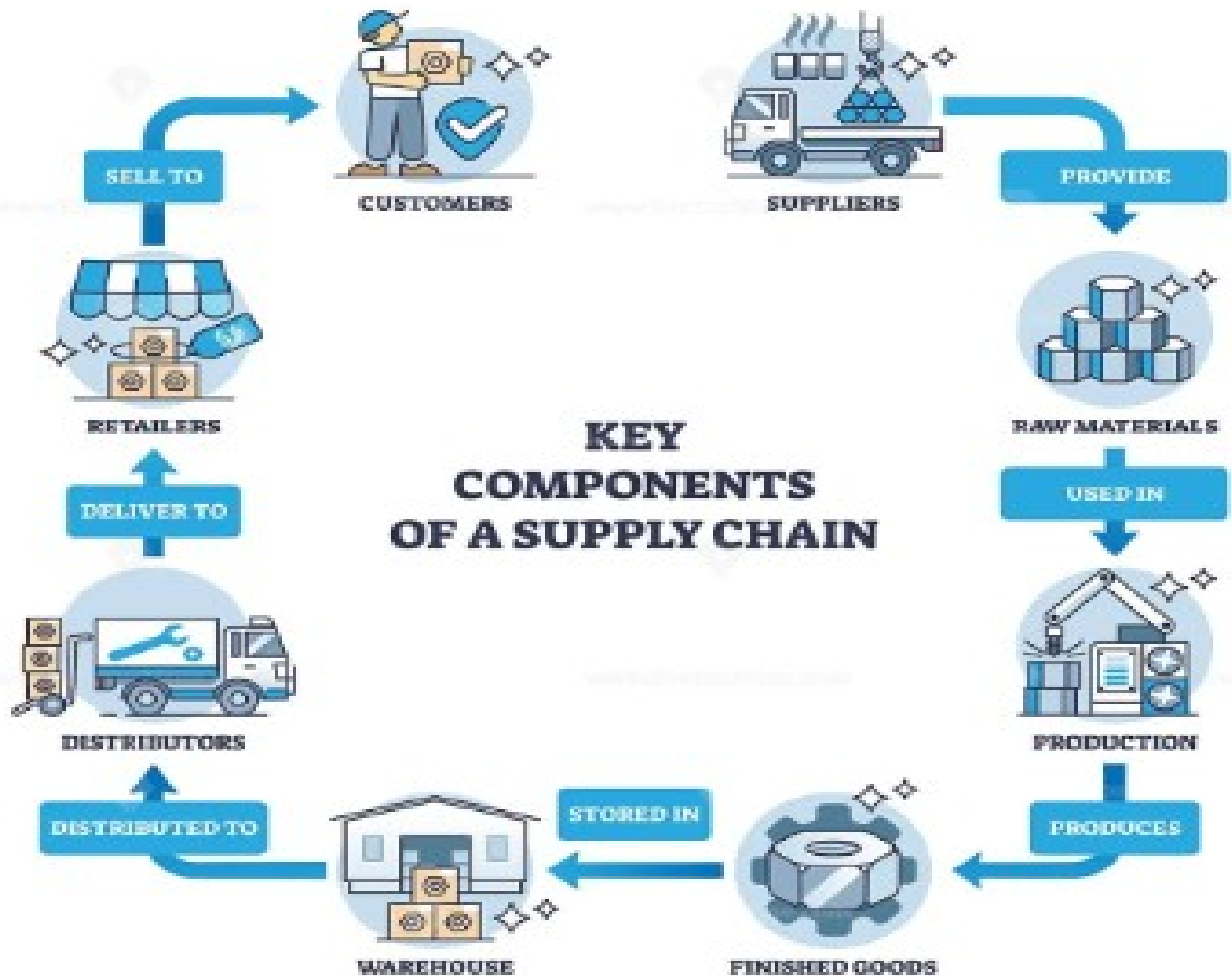
- **Core Competency in Supply Chains**
- **Core competencies** refer to the unique strengths or capabilities that give a company a competitive advantage in the market.
- In the context of supply chains, these competencies are often related to **specialized knowledge** or **technologies** that enable a company to perform tasks better than its competitors.
- **In Healthcare:**
 - Healthcare providers might focus on **advanced logistics management**
 - Pharmaceutical companies might emphasize their **expertise in manufacturing** and ensuring the **quality and safety** of drugs.
 - **Technology and data management:** Companies might use **real-time tracking** and **predictive analytics** to manage **inventory** and **demand forecasting**.

Supply Chain Map

- A **supply chain map** is a **visual representation** that illustrates the flow of **materials**, **information**, and **finances** across the supply chain.
- It helps to show how goods and services move from the **initial suppliers** to the **final customers** (patients in healthcare), as well as the key stakeholders involved at each step.
- **Purpose:**
 - **Understanding relationships** between different entities in the supply chain.
 - **Identifying key touch points** where critical actions (e.g., inventory management, procurement, logistics) occur.
 - Offering **visibility** into the processes of procurement, production, transportation, and distribution.

Supply Chain Management Cycle





Main Components of a Healthcare Supply Chain Map

- **Manufacturers:**
The entities responsible for producing **medicines, medical devices,** and other healthcare products.
- **Distributors:**
Organizations that manage the **transportation, warehousing,** and **delivery** of products from manufacturers to healthcare providers.
- **Healthcare Providers** (e.g., Hospitals, Clinics):
Healthcare institutions that **administer** products to **patients**. These include hospitals, clinics, pharmacies, and long-term care facilities.
- **Patients:**
The ultimate recipients of the healthcare products and services.
- **Regulatory Bodies:**
Government or industry organizations that oversee the **compliance, safety,** and **quality** standards of healthcare products and processes.

Why Create Supply Chain Maps?

- **Identify Inefficiencies and Bottlenecks:**
Mapping the supply chain allows stakeholders to identify areas where there are delays, inefficiencies, or gaps in the process.
- **Support Better Decision-Making:**
Supply chain maps provide **visibility** into how goods and services move. This helps organizations make **informed decisions** regarding procurement, production planning, and inventory management.
- **Optimize Logistics:**
Understanding the logistics flow from supplier to patient helps to **streamline transportation**, reduce lead times, and improve **cost-efficiency**.
- **Improve Risk Management:**
A well-developed map helps to anticipate potential risks in the supply chain (e.g., natural disasters, regulatory changes) and develop mitigation strategies.

Globalization of Supply Chains

- The globalization of trade has significantly transformed healthcare supply chains.
- This has increased efficiency and competition, but also created new challenges and dependencies across borders.
- **Examples in Healthcare:**
 - **Sourcing Medical Devices:** Many medical devices (e.g., **MRI machines, surgical instruments**) are manufactured in countries like **China, Germany, and the US**, but used globally. This international supply chain connects **designers, manufacturers, and distributors** worldwide.
 - **Sourcing Drugs:** Countries often rely on pharmaceuticals produced in countries like **India and China**. For example, **generic medicines** for various diseases are commonly produced in India and exported to other markets, including the US and European countries.
 - **Raw Materials:** Certain raw materials (e.g., **active pharmaceutical ingredients or medical-grade plastics**) are sourced globally and form the basis for manufacturing healthcare products in different regions.

Globalization Benefits:

- **Cost-effectiveness:** Global trade allows healthcare systems to benefit from **lower production costs**, access to diverse suppliers, and economies of scale.
- **Faster innovation:** Collaborations and international partnerships allow for the rapid development and distribution of new medical technologies and treatments.

Challenges of Global Supply Chains

- **Lead Time and Inventory Management:**
 - The longer **lead times** associated with global sourcing increase the complexity of managing **inventory** levels.
 - Delays due to customs, transportation issues, or demand fluctuations can create supply **disruptions**.
- **Political Instability, Trade Tariffs, and Regulations:**
 - **Political instability** in key sourcing countries (e.g., natural disasters, conflicts) can disrupt the supply of essential medical products.
 - **Trade tariffs** and **regulatory differences** across borders can increase costs or cause delays in supply delivery.
 - **Changing regulations** in different countries (e.g., import/export restrictions, FDA approval processes) can complicate the flow of goods, particularly medical products, across borders.

Thank You!

Quiz 2

1. Try to define the following two terminologies in Supply Chain Management
 - Postponement Strategy
 - Bullwhip effect